

# Permutations — Solutions

## Warm-up

### Question 1:

The number of ways to arrange  $n$  distinct objects in a line is given by  $n!$ .

$$\begin{aligned}\text{Number of ways} &= 5! \\ &= 5 \times 4 \times 3 \times 2 \times 1 \\ &= 120\end{aligned}$$

### Question 2:

The word **LOGIC** contains 5 distinct letters. The number of arrangements is  $5!$ .

$$\begin{aligned}\text{Arrangements} &= 5! \\ &= 120\end{aligned}$$

### Question 3:

The number of ways to arrange 6 distinct books is  $6!$ .

$$\begin{aligned}\text{Ways} &= 6! \\ &= 720\end{aligned}$$

### Question 4:

The word **CIDER** contains 5 distinct letters. The number of arrangements is  $5!$ .

$$\begin{aligned}\text{Ways} &= 5! \\ &= 120\end{aligned}$$

### Question 5:

There are 5 positions in the line. The first position is fixed for the specific person.

$$\begin{aligned}\text{Ways} &= 1 \times 4! \\ &= 1 \times 24 \\ &= 24\end{aligned}$$

### Question 6:

The word **APPLE** contains 5 letters, where the letter 'P' is repeated twice.

$$\begin{aligned}
 \text{Distinct arrangements} &= \frac{5!}{2!} \\
 &= \frac{120}{2} \\
 &= 60
 \end{aligned}$$

**Question 7:**

The number of ways to arrange  $n$  distinct objects around a circular table is  $(n - 1)!$ .

$$\begin{aligned}
 \text{Ways} &= (4 - 1)! \\
 &= 3! \\
 &= 6
 \end{aligned}$$

**Question 8:**

We need to form a 3-digit number using 5 available digits without repetition. This is a permutation of 5 objects taken 3 at a time.

$$\begin{aligned}
 \text{Ways} &= P(5, 3) \\
 &= \frac{5!}{(5 - 3)!} \\
 &= \frac{120}{2} \\
 &= 60
 \end{aligned}$$

**Question 9:**

Treat the 3 people who must sit together as a single block. This leaves us with  $6 - 3 + 1 = 4$  entities to arrange.

$$\begin{aligned}
 \text{Ways to arrange entities} &= 4! \\
 \text{Ways to arrange people within block} &= 3! \\
 \text{Total ways} &= 4! \times 3! \\
 &= 24 \times 6 \\
 &= 144
 \end{aligned}$$

**Question 10:**

The word **BANANA** contains 6 letters: 1 B, 3 A's, and 2 N's.

$$\begin{aligned}
\text{Arrangements} &= \frac{6!}{3!2!1!} \\
&= \frac{720}{6 \times 2} \\
&= \frac{720}{12} \\
&= 60
\end{aligned}$$

**Question 11:**

The license plate has 2 letters (26 options each) and 3 digits (10 options each), with repetition allowed.

$$\begin{aligned}
\text{Total plates} &= 26^2 \times 10^3 \\
&= 676 \times 1000 \\
&= 676,000
\end{aligned}$$

## Skill-building

**Question 1:**

The word **KITCHEN** has 7 letters. Vowels are I, E and consonants are K, T, C, H, N.

Treat the vowels as one block. We now have 5 consonants + 1 block = 6 entities.

$$\begin{aligned}
\text{Ways to arrange entities} &= 6! \\
\text{Ways to arrange vowels within block} &= 2! \\
\text{Total ways} &= 6! \times 2! \\
&= 720 \times 2 \\
&= 1440
\end{aligned}$$

**Question 2:**

Total arrangements of 8 people is  $8!$ . We subtract the cases where the 2 particular people sit together.

If they sit together, treat them as one block. We have  $8 - 2 + 1 = 7$  entities.

$$\begin{aligned}
\text{Ways they sit together} &= 7! \times 2! \\
\text{Total ways} &= 8! - (7! \times 2!) \\
&= 40320 - (5040 \times 2) \\
&= 40320 - 10080 \\
&= 30240
\end{aligned}$$

**Question 3:**

The word **MISSISSIPPI** has 11 letters: 1 M, 4 I's, 4 S's, 2 P's.

$$\begin{aligned}
 \text{Arrangements} &= \frac{11!}{4!4!2!} \\
 &= \frac{39916800}{24 \times 24 \times 2} \\
 &= \frac{39916800}{1152} \\
 &= 34650
 \end{aligned}$$

**Question 4:**

Seven people are at a round table. Treat A and B as one block. This gives us  $7 - 2 + 1 = 6$  entities to arrange circularly.

$$\begin{aligned}
 \text{Circular arrangements of 6 entities} &= (6 - 1)! = 5! \\
 \text{Ways A and B can swap} &= 2! \\
 \text{Total ways} &= 5! \times 2! \\
 &= 120 \times 2 \\
 &= 240
 \end{aligned}$$

**Question 5:**

A 4-digit number is even if it ends in 2, 4, or 6 (3 options). The remaining 3 positions are filled from the remaining 5 digits.

$$\begin{aligned}
 \text{Ways to choose last digit} &= 3 \\
 \text{Ways to fill remaining 3 positions} &= P(5, 3) \\
 \text{Total ways} &= 3 \times (5 \times 4 \times 3) \\
 &= 3 \times 60 \\
 &= 180
 \end{aligned}$$

**Question 6:**

The word **CHORUS** has 6 letters: Consonants C, H, R, S and Vowels O, U.

There are 4 consonants available for the first position and 3 for the last position.

$$\begin{aligned}
 \text{Ways to pick first and last letters} &= 4 \times 3 \\
 \text{Ways to arrange remaining 4 letters} &= 4! \\
 \text{Total ways} &= (4 \times 3) \times 4! \\
 &= 12 \times 24 \\
 &= 288
 \end{aligned}$$

**Question 7:**

Two patterns are possible: BGBGBGBGBG or GBGBGBGBGB.

$$\text{Ways for BGBG...} = 5! \times 5!$$

$$\text{Ways for GBGB...} = 5! \times 5!$$

$$\begin{aligned} \text{Total ways} &= 2 \times (5! \times 5!) \\ &= 2 \times (120 \times 120) \\ &= 2 \times 14400 \\ &= 28800 \end{aligned}$$

**Question 8:**

First, arrange the 3 people who are NOT separated (the other 3 people). Then, place the 3 specific people in the gaps between them.

$$\text{Ways to arrange the other 3 people} = 3!$$

$$\text{Number of gaps available} = 4$$

$$\text{Ways to place 3 people in 4 gaps} = P(4, 3)$$

$$\begin{aligned} \text{Total ways} &= 3! \times P(4, 3) \\ &= 6 \times (4 \times 3 \times 2) \\ &= 6 \times 24 \\ &= 144 \end{aligned}$$

**Question 9:**

Five people at a circular table. Treat the 2 particular people as one block. This gives  $5 - 2 + 1 = 4$  entities.

$$\text{Circular arrangements of 4 entities} = (4 - 1)! = 3!$$

$$\text{Ways the 2 people can swap} = 2!$$

$$\begin{aligned} \text{Total ways} &= 3! \times 2! \\ &= 6 \times 2 \\ &= 12 \end{aligned}$$

**Question 10:**

The word **CHORUS** has 4 consonants C, H, R, S.

$$\text{Ways to pick the first letter} = 4$$

$$\text{Ways to arrange remaining 5 letters} = 5!$$

$$\begin{aligned} \text{Total ways} &= 4 \times 5! \\ &= 4 \times 120 \\ &= 480 \end{aligned}$$

## Easier Exam Questions

### Question 1:

The number of ways to arrange 10 people in a straight line is  $10!$ .

$$\begin{aligned}\text{Ways} &= 10! \\ &= 3,628,800\end{aligned}$$

### Question 2:

Peter, Jeffery, and Kevin must be at the beginning. They can be arranged among themselves in  $3!$  ways. The remaining 4 boys can be arranged in  $4!$  ways.

$$\begin{aligned}\text{Ways} &= 3! \times 4! \\ \text{This matches option (C).}\end{aligned}$$

### Question 3:

The plate has 3 letters (26 options each) and 2 numbers (10 options each).

$$\begin{aligned}\text{Ways} &= 26^3 \times 10^2 \\ &= 17576 \times 100 \\ &= 1,757,600\end{aligned}$$

### Question 4(i):

The word **POSSIBILITY** has 11 letters: 1 P, 1 O, 1 P, 1 O, 2 S's, 2 I's, 1 L, 1 T, 1 Y. Let's re-count: P(1), O(1), S(2), I(2), B(1), L(1), T(1), Y(1). Total = 10 letters.

$$\begin{aligned}\text{Arrangements} &= \frac{10!}{2!2!} \\ &= \frac{3628800}{4} \\ &= 907,200\end{aligned}$$

### Question 4(ii):

Treat the two S's as one block. We now have  $10 - 2 + 1 = 9$  entities. The letters are P, O, (SS), I, I, B, L, T, Y.

$$\begin{aligned}\text{Arrangements} &= \frac{9!}{2!} \\ &= \frac{362880}{2} \\ &= 181,440\end{aligned}$$

**Question 4(iii):**

Fix the S's at the first and last positions. We arrange the remaining 8 letters: P, O, I, I, B, L, T, Y.

$$\begin{aligned}\text{Arrangements} &= \frac{8!}{2!} \\ &= \frac{40320}{2} \\ &= 20,160\end{aligned}$$

**Question 5(i):**

The word **COMMITTEE** has 9 letters: 2 C's (No, 1 C), 1 C, 2 O's (No, 1 O), 1 O, 2 M's, 2 T's, 2 E's, 1 I. Total = 9 letters. Let's re-count: C(1), O(1), M(2), I(1), T(2), E(2). Total = 9 letters.

$$\begin{aligned}\text{Arrangements} &= \frac{9!}{2!2!2!} \\ &= \frac{362880}{8} \\ &= 45,360\end{aligned}$$

**Question 5(ii):**

Treat the two E's as one block. We now have  $9 - 2 + 1 = 8$  entities: C, O, M, M, I, T, T, (EE).

$$\begin{aligned}\text{Arrangements} &= \frac{8!}{2!2!} \\ &= \frac{40320}{4} \\ &= 10,080\end{aligned}$$

**Question 6:**

This is identical to Question 2. Peter, Jeffery, and Kevin must be at the beginning. They can be arranged in  $3!$  ways. The remaining 4 boys can be arranged in  $4!$  ways.

$$\text{Ways} = 3! \times 4!$$

This matches option (C).

**Question 7:**

The word **MATHS** has 5 distinct letters.

$$\begin{aligned}\text{Arrangements} &= 5! \\ &= 120\end{aligned}$$

**Question 8(i):**

The word **DELETED** has 7 letters: 3 E's, 2 D's, 1 L, 1 T.

$$\begin{aligned}
 \text{Total arrangements} &= \frac{7!}{3!2!} \\
 &= \frac{5040}{6 \times 2} \\
 &= \frac{5040}{12} \\
 &= 420
 \end{aligned}$$

**Question 8(ii):**

To find where D's are separated, we subtract the cases where D's are together from the total.

If D's are together, treat them as one block (DD). We have  $7 - 2 + 1 = 6$  entities: (DD), E, E, E, L, T.

$$\begin{aligned}
 \text{Ways D's are together} &= \frac{6!}{3!} \\
 &= \frac{720}{6} \\
 &= 120
 \end{aligned}$$

$$\begin{aligned}
 \text{Ways D's are separated} &= 420 - 120 \\
 &= 300
 \end{aligned}$$

**Question 9(i):**

Total people =  $5 + 6 = 11$ .

$$\begin{aligned}
 \text{Ways} &= 11! \\
 &= 39,916,800
 \end{aligned}$$

**Question 9(ii):**

Treat the 5 boys as one block. We now have  $11 - 5 + 1 = 7$  entities (the block of boys and 6 girls).

$$\begin{aligned}
 \text{Ways to arrange entities} &= 7! \\
 \text{Ways to arrange boys within block} &= 5! \\
 \text{Total ways} &= 7! \times 5! \\
 &= 5040 \times 120 \\
 &= 604,800
 \end{aligned}$$

**Question 10:**

The word **DELETED** has 7 letters. We are making 7-letter arrangements, so we use all letters.

$$\begin{aligned}
 \text{Ways} &= \frac{7!}{3!2!} \\
 &= \frac{5040}{12} \\
 &= 420
 \end{aligned}$$

This matches option (A).

**Question 11:**

The word **EMBEZZLED** has 9 letters: 3 E's, 2 Z's, 1 M, 1 B, 1 L, 1 D.

$$\begin{aligned}
 \text{Arrangements} &= \frac{9!}{3!2!} \\
 &= \frac{362880}{6 \times 2} \\
 &= \frac{362880}{12} \\
 &= 30,240
 \end{aligned}$$

This matches option (C).

## Harder Exam Questions

**Question 1:**

Total circular arrangements of 7 people is  $(7 - 1)! = 6! = 720$ .

If Preksha and Chloe sit together, treat them as one block. This gives  $7 - 2 + 1 = 6$  entities.

$$\begin{aligned}
 \text{Ways they sit together} &= (6 - 1)! \times 2! \\
 &= 5! \times 2 \\
 &= 120 \times 2 \\
 &= 240
 \end{aligned}$$

$$\begin{aligned}
 \text{Ways they do NOT sit together} &= 720 - 240 \\
 &= 480
 \end{aligned}$$

This matches option (A).

**Question 2:**

First, arrange the 4 boys in a circle. This can be done in  $(4 - 1)! = 3! = 6$  ways.

There are 4 gaps between the boys. We need to place 2 girls in these 4 gaps.

$$\begin{aligned}
 \text{Ways to place girls} &= P(4, 2) \\
 &= 4 \times 3 \\
 &= 12
 \end{aligned}$$

$$\begin{aligned}\text{Total arrangements} &= 6 \times 12 \\ &= 72\end{aligned}$$

This matches option (D).

**Question 3:**

In any arrangement of 6 people, the oldest person (O) and the youngest person (Y) will be in some relative order. There are only two possibilities: O is before Y, or Y is before O.

Due to symmetry, in exactly half of the  $6!$  total arrangements, the oldest person will be in front of the youngest.

$$\begin{aligned}\text{Ways} &= \frac{6!}{2} \\ &= \frac{720}{2} \\ &= 360\end{aligned}$$

This matches option (B).

**Question 4:**

The word **TROOPS** has 6 letters: 2 O's, 1 T, 1 R, 1 P, 1 S.

$$\text{Total arrangements} = \frac{6!}{2!} = \frac{720}{2} = 360.$$

If the O's are together, treat them as one block (OO). We have  $6 - 2 + 1 = 5$  entities: (OO), T, R, P, S.

$$\begin{aligned}\text{Ways O's are together} &= 5! \\ &= 120\end{aligned}$$

$$\begin{aligned}\text{Ways O's are separated} &= 360 - 120 \\ &= 240\end{aligned}$$

This matches option (B).

**Question 5a:**

The word **DEDUCED** has 7 letters: 3 D's, 2 E's, 1 U, 1 C, 1 D

$$\begin{aligned}\text{Arrangements} &= \frac{7!}{3!2!} \\ &= \frac{5040}{6 \times 2} \\ &= 420\end{aligned}$$

**Question 5b:**

We need to form numbers greater than 3000 using digits 1, 3, 5, 7, 9 without repetition. This means the number can be 4 digits or 5 digits.

Case 1: 4-digit numbers. The first digit must be 3, 5, 7, or 9 (4 options). The remaining 3 positions are filled from the remaining 4 digits.

$$\begin{aligned}\text{Ways (4-digit)} &= 4 \times P(4, 3) \\ &= 4 \times (4 \times 3 \times 2) \\ &= 4 \times 24 \\ &= 96\end{aligned}$$

Case 2: 5-digit numbers. Any arrangement of the 5 digits will be greater than 3000.

$$\begin{aligned}\text{Ways (5-digit)} &= 5! \\ &= 120\end{aligned}$$

$$\begin{aligned}\text{Total ways} &= 96 + 120 \\ &= 216\end{aligned}$$

**Question 5d:**

Total people = 8 (2 hosts, 6 guests). Total circular arrangements =  $(8 - 1)! = 7! = 5040$ .

If the hosts sit together, treat them as one block. This gives  $8 - 2 + 1 = 7$  entities.

$$\begin{aligned}\text{Ways hosts sit together} &= (7 - 1)! \times 2! \\ &= 6! \times 2 \\ &= 720 \times 2 \\ &= 1440\end{aligned}$$

$$\begin{aligned}\text{Ways hosts are separated} &= 5040 - 1440 \\ &= 3600\end{aligned}$$

**Question 6:**

The word **ROAMING** has 7 distinct letters. Vowels are O, A, I and consonants are R, M, N, G.

The word must end in a vowel. There are 3 options for the last position.

$$\begin{aligned}\text{Ways to pick last letter} &= 3 \\ \text{Ways to arrange remaining 6 letters} &= 6! \\ \text{Total ways} &= 3 \times 6! \\ &= 3 \times 720 \\ &= 2160\end{aligned}$$

This matches option (B).

**Question 7(i):**

The word **STREET** has 6 letters: 2 E's, 2 T's, 1 S, 1 R.

$$\text{Total arrangements} = \frac{6!}{2!2!} = \frac{720}{4} = 180.$$

If the two E's are together, treat them as one block (EE). We have  $6 - 2 + 1 = 5$  entities: (EE), S, T, T, R.

$$\begin{aligned}\text{Ways E's are together} &= \frac{5!}{2!} \\ &= \frac{120}{2} \\ &= 60\end{aligned}$$

$$\begin{aligned}\text{Probability} &= \frac{60}{180} \\ &= \frac{1}{3}\end{aligned}$$

**Question 7(ii):**

If the E's and T's are together in one group, treat the group E, E, T, T as one block. We have  $6 - 4 + 1 = 3$  entities: (EETT), S, R.

$$\begin{aligned}\text{Ways to arrange entities} &= 3! \\ \text{Ways to arrange letters within block} &= \frac{4!}{2!2!} \\ &= \frac{24}{4} \\ &= 6\end{aligned}$$

$$\begin{aligned}\text{Total ways} &= 3! \times 6 \\ &= 6 \times 6 \\ &= 36\end{aligned}$$

$$\begin{aligned}\text{Probability} &= \frac{36}{180} \\ &= \frac{1}{5}\end{aligned}$$

**Question 8:**

Eight chairs are around a circular table. No student is directly opposite a teacher. This means if a teacher is at position  $i$ , a student cannot be at position  $i + 4$ .

This implies that for every pair of opposite seats, either both seats must be occupied by teachers or both seats must be occupied by students.

There are 4 pairs of opposite seats. We must choose 2 pairs for teachers and 2 pairs for students.

$$\begin{aligned}\text{Ways to choose pairs} &= \binom{4}{2} \\ &= 6\end{aligned}$$

$$\text{Ways to arrange teachers in their 4 seats} = 4!$$

$$\text{Ways to arrange students in their 4 seats} = 4!$$

$$\begin{aligned}\text{Total ways (linear)} &= 6 \times 4! \times 4! \\ &= 6 \times 24 \times 24 \\ &= 3456\end{aligned}$$

Since it is a circular table, we divide by the number of rotations (8).

$$\begin{aligned}\text{Circular ways} &= \frac{3456}{8} \\ &= 432\end{aligned}$$

This matches option (C).

### Question 9:

The word **REPETITION** has 10 letters: 2 E's, 2 T's, 2 I's, 1 R, 1 P, 1 O, 1 N. Vowels are E, E, I, I, O (5) and consonants are R, P, T, T, N (5).

Two patterns are possible: VCVVCVCVCVC or CVCVCVCVCV.

$$\begin{aligned}\text{Ways to arrange vowels} &= \frac{5!}{2!2!} \\ &= \frac{120}{4} \\ &= 30\end{aligned}$$

$$\begin{aligned}\text{Ways to arrange consonants} &= \frac{5!}{2!} \\ &= \frac{120}{2} \\ &= 60\end{aligned}$$

$$\begin{aligned}\text{Total ways} &= 2 \times (30 \times 60) \\ &= 2 \times 1800 \\ &= 3600\end{aligned}$$

This matches option (B).

**Question 10:**

Five boys and two girls are to be arranged in a row. Treat the two girls as one block.

$$\begin{aligned}\text{Ways to arrange entities} &= (5 + 1)! = 6! \\ \text{Ways to arrange girls within block} &= 2! \\ \text{Total ways} &= 6! \times 2! \\ &= 720 \times 2 \\ &= 1440\end{aligned}$$

This matches option (B).

**Question 11:**

The word **QUACKED** has 7 letters: 4 consonants Q, C, K, D and 3 vowels U, A, E.

To alternate, the pattern must be CVCVCVC.

$$\begin{aligned}\text{Ways to arrange consonants} &= 4! \\ \text{Ways to arrange vowels} &= 3! \\ \text{Total ways} &= 4! \times 3! \\ &= 24 \times 6 \\ &= 144\end{aligned}$$

This matches option (C).

**Question 12:**

Five black dresses (B) and four white dresses (W). To alternate, the pattern must be BWBWB-WBWB.

$$\begin{aligned}\text{Ways to arrange black dresses} &= 5! \\ \text{Ways to arrange white dresses} &= 4! \\ \text{Total ways} &= 5! \times 4! \\ \text{This matches option (A).}\end{aligned}$$

**Question 13:**

Eight friends are in a row. Alice (A) and Bob (B) must be in aisle seats (the two ends).

$$\begin{aligned}\text{Ways to place A and B} &= 2! \\ \text{Remaining seats} &= 6 \\ \text{Remaining people} &= 6\end{aligned}$$

Condition: Corinne (C) sits the same number of seats away from Douglas (D) as she does from Ellen (E).

This means C must be the midpoint of D and E. For the 6 internal seats (positions 2, 3, 4, 5, 6, 7), the possible sets of positions for D, C, E are: - (2, 3, 4) - (3, 4, 5) - (4, 5, 6) - (5, 6, 7) - (2, 4, 6) - (3, 5, 7)

There are 6 such sets of positions. For each set, C must be in the middle, and D and E can swap.

$$\begin{aligned}\text{Ways to arrange D, C, E} &= 6 \times 2! \\ &= 12\end{aligned}$$

$$\begin{aligned}\text{Ways to arrange the other 3 friends} &= 3! \\ &= 6\end{aligned}$$

$$\begin{aligned}\text{Total ways} &= 2! \times 12 \times 6 \\ &= 2 \times 12 \times 6 \\ &= 144\end{aligned}$$